VNAV Lab 4 - Individual Assignment

Single-Segment and Multi-Segment Trajectory Optimization

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Deliverable 1: Single-Segment Trajectory Optimization

We minimize the squared velocity over the unit interval:

$$\text{minimize} \quad J(P)=\int_0^1 [P'(t)]^2 dt, \quad \text{subject to} \quad P(0)=0, \quad P(1)=1.$$

1. Polynomial of degree 1

Let $P(t)=p_1 t$. The terminal condition $P(1)=1$ gives $p_1=1$. Thus $P'(t)=1$ and

$$J(P)=\int_0^1 1^2 dt=1.$$

> Result. The optimal trajectory is $P^*(t)=t$, and the optimal cost is $J^*=1$.

2. Polynomial of degree 2

Let $P(t)=p_1 t+p_2 t^2$. Since $P'(t)=p_1+2p_2 t$,

$$J(P)=\int_0^1 (p_1+2p_2 t)^2 dt=p_1^2+2p_1 p_2+{}^4/_3\, p_2^2.$$

(a) With $p=[p_1,p_2]^T$, the cost is $J=p^T Q p$, where

$$Q = \begin{bmatrix} 1 & 1 \\ 1 & 4/3 \end{bmatrix}$$

(b) The terminal constraint is $p_1+p_2=1$. Therefore

$$A=[1 \quad 1], \quad b=[1], \quad Ap=b.$$

(c) Substituting $p_1=1-p_2$ into the cost gives

$$J(p_2)=1+{}^1/_3\, p_2^2.$$

The unique minimum occurs at $p_2=0$, so $p_1=1$.

> Result. $P^*(t)=t$ and $J^*=1$. Allowing a quadratic term does not reduce the cost.

3. Polynomial of degree 3

Let $P(t)=p_1t+p_2t^2+p_3t^3$. For $i,j\in\{1,2,3\}$,

$$Q_{ij}=\int_0^1(it^{i-1})(jt^{j-1})dt=\frac{ij}{i+j-1}.$$

(a) Hence the QP matrices are

$$Q=\begin{bmatrix}1 & 1 & 1\\ 1 & 4/3 & 3/2\\ 1 & 3/2 & 9/5\end{bmatrix}, \qquad A=[1\quad 1\quad 1], \quad b=[1].$$

(b) By the Cauchy-Schwarz inequality,

$$J(P)=\int_0^1[P'(t)]^2dt\geq\left(\int_0^1P'(t)dt\right)^2=[P(1)-P(0)]^2=1.$$

Equality holds exactly when $P'(t)$ is constant. The endpoint conditions then imply $P'(t)=1$, so $p_1=1$ and $p_2=p_3=0$.

> Result. $P^*(t)=t$ and $J^*=1$. This agrees with the Euler-Lagrange result: for $r=1$, $P^{(2)}(t)=0$, so the extremal has degree $2r-1=1$.

4. Additional constraint $P'(1)=-2$

The straight line $P(t)=t$ is no longer feasible. We solve the equality-constrained QP separately for degrees 2 and 3.

(a) Degree 2

The endpoint position and terminal velocity give

$$p_1+p_2=1, \quad p_1+2p_2=-2.$$

Thus $p_2=-3$ and $p_1=4$. Substitution into the quadratic cost gives

$$J=4^2+2(4)(-3)+\frac{4}{3}(-3)^2=4.$$

> Result. $P^*(t)=4t-3t^2$, with $J^*=4$.

4(b). Degree 3

The two equality constraints are

$$p_1+p_2+p_3=1, \quad p_1+2p_2+3p_3=-2.$$

Set $x=p_3$. Solving the constraints gives $p_2=-3-2x$ and $p_1=4+x$. The cost becomes

$$J(x)={}^{2}/_{15}x^2+x+4, \quad J'(x)={}^{4}/_{15}x+1.$$

The stationary point is $x=-15/4$. Because $J''(x)=4/15>0$, it is the unique minimum. Therefore

$$p_1={}^{1}/_{4}, \quad p_2={}^{9}/_{2}, \quad p_3=-{}^{15}/_{4}.$$

$$P^*(t)={}^{1}/_{4}t+{}^{9}/_{2}t^2-{}^{15}/_{4}t^3.$$

The constraints are satisfied:

$$P^*(0)=0, \quad P^*(1)=1, \quad P^{*'}(1)=-2.$$

Finally,

$$J^*={}^{2}/_{15}({}^{225}/_{16})-{}^{15}/_{4}+4={}^{17}/_{8}=2.125.$$

> Result. The cubic solution has cost 17/8, which is lower than the quadratic cost 4.

Deliverable 2: Multi-Segment Trajectory Optimization

Let $t_0<t_1<\ldots<t_k$ define k trajectory segments. Segment i is represented by $P_i(t)$, and the prescribed scalar waypoints are $\bar{x}_0,\bar{x}_1,\ldots,\bar{x}_k$. All counts below are for one scalar coordinate.

1. Minimum snap (r=4)

For $L=[P^{(4)}(t)]^2$, the Euler-Lagrange equation is $P^{(8)}(t)=0$. Hence each segment has degree 7 and therefore 8 coefficients. The k-segment trajectory has 8k unknown coefficients.

Constraint count for minimum snap

Type	Constraints	Count
Waypoint	For every segment i: $P_i(t_{i-1})=\bar{x}_{i-1}$ and $P_i(t_i)=\bar{x}_i$. An internal waypoint is imposed on both adjacent segments.	2k
Free derivative	At every internal waypoint t_i, enforce $P_i^{(j)}(t_i)=P_{i+1}^{(j)}(t_i)$ for j=1,...,6.	6(k-1)
Fixed derivative	At t_0 and t_k, prescribe derivatives j=1,2,3: velocity, acceleration, and jerk. For a rest-to-rest trajectory these values are zero.	6
Total		2k+6(k-1)+6 =8k

At an internal waypoint t_i, the equations are

$$P_i(t_i)=\bar{x}_i, \quad P_{i+1}(t_i)=\bar{x}_i, \quad P_i^{(j)}(t_i)=P_{i+1}^{(j)}(t_i), \quad j=1,\ldots,6.$$

Result. The required constraints are 2k waypoint constraints, 6(k-1) free-derivative continuity constraints, and 6 fixed endpoint-derivative constraints, for a total of 8k.

2. General r-th derivative over k segments

For $L=[P^{(r)}(t)]^2$, the generalized Euler-Lagrange equation reduces to $P^{(2r)}(t)=0$. Each segment therefore has degree 2r-1 and 2r coefficients, giving 2rk unknown coefficients in total.

Type	Orders and locations	Count
Waypoint	Both endpoints of each of the k segments	2k
Free derivative	Continuity at each of the k-1 internal points for derivative orders j=1,...,2r-2	(2r-2)(k-1)
Fixed derivative	At both outer endpoints, prescribe derivative orders j=1,...,r-1	2(r-1)
Total		2rk

Indeed, the count closes exactly:

$$2k+(2r-2)(k-1)+2(r-1)=2rk.$$

Result. For the general problem, use degree 2r-1 on every segment, with 2k waypoint constraints, (2r-2)(k-1) free-derivative constraints, and 2(r-1) fixed endpoint-derivative constraints.

References
[1] MIT VNAV, Lab 4 Exercises (2023), https://vnav.mit.edu/labs_2023/lab4/exercises.html
[2] MIT VNAV, Lecture 9: Trajectory Optimization I, https://vnav.mit.edu/material/09-TrajectoryOptimization1-notes.pdf